LESSON 8.4

Write and Represent Inequalities

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write inequalities to represent real-world situations.

WHY THIS MATTERS Detective Navarro is narrowing down suspects. A witness says the suspect is at least 18 years old. The detective writes the inequality a ≥ 18, where a is the suspect's age. How does this help eliminate suspects?

REMEMBER

Write and Represent Inequalities

Writing & Representing Inequalities. Formula: < (less) | > (greater) | ≤ (at most / ≤) | ≥ (at least / ≥) An inequality compares two amounts using <, >, ≤, or ≥. I use one when many values are allowed, not just one. I pick a variable and match the words to the right symbol.

1. 'At least', 'minimum', 'no less than' translate to ≥ (greater than or equal to).
2. 'At most', 'maximum', 'no more than' translate to ≤ (less than or equal to).
3. 'More than' is >; 'Fewer than' / 'Less than' is <

Words: **Inequality *(Desigualdad)*** **Greater than *(Mayor que)*** **Less than *(Menor que)*** **At least / At most *(Al menos / A lo más)*** **Variable *(Variable)***

1 **Circle the letter of the best answer. Show how you know.**

Solve: 6x = 42

(A) x = 7
(B) x = 36
(C) x = 48
(D) x = 252

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Solve: n / 3 = 8

(A) n = 3
(B) n = 5
(C) n = 11
(D) n = 24

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Solve: 9m = 63

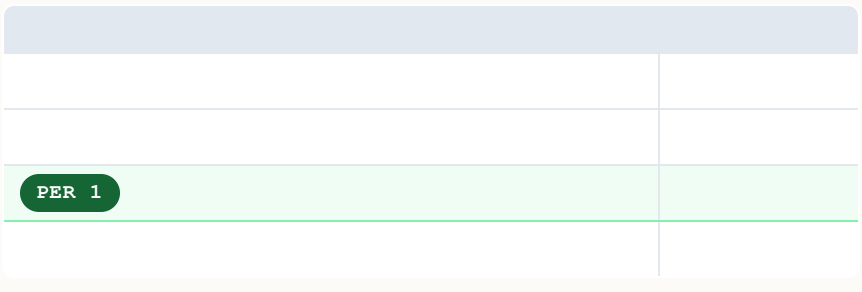
(A) m = 7
(B) m = 54
(C) m = 72
(D) m = 567

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A detective knows the getaway car was traveling at most 65 miles per hour on the highway. The inequality s ≤ 65 represents the car's speed s. Which speed is NOT possible for the getaway car?

(A) 45 mph
(B) 60 mph
(C) 65 mph
(D) 70 mph



1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

5 **Circle the letter of the best answer. Show how you know.**

A detective knows the getaway car was traveling at most 65 miles per hour on the highway. The inequality $s \leq 65$ represents the car's speed s. Does $s \leq 65$ include 65 itself?

(A) No — it must be strictly less
(B) Yes — the 'or equal to' part includes 65
(C) Only if the car stops
(D) Only on a highway

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A detective knows the getaway car was traveling at most 65 miles per hour on the highway. The inequality $s \leq 65$ represents the car's speed s. Which inequality means 'at most 65'?

(A) $s \leq 65$
(B) $s < 65$
(C) $s \geq 65$
(D) $s > 65$

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do the words 'at least' and 'at most' change the inequality symbol compared to 'more than' and 'fewer than'?

SENTENCE STARTER 'At least' uses ≥ instead of > because it means the value ___ be equal to the number, while 'more than' means it must be ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 8.4

Write and Represent Inequalities

Set B · Version B · Second form · core practice

MASTERY CHECK
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Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write inequalities to represent real-world situations.

WHY THIS MATTERS Detective Navarro is narrowing down suspects. A witness says the suspect is at least 18 years old. The detective writes the inequality $a \geq 18$, where a is the suspect's age. How does this help eliminate suspects?

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Each case clue describes a boundary. Drag every scenario into the inequality symbol it matches.

Groups: **> (greater than)** **< (less than)** **≥ (greater than or equal to)** **≤ (less than or equal to)**

_____ The getaway driver was going more than 70 mph

_____ More than 6 fingerprints were on the safe

_____ The ransom note was written under 3 minutes ago

_____ Fewer than 9 cars left the parking lot

_____ At least 30 dollars in coins was stolen

_____ No fewer than 4 witnesses saw the suspect

_____ At most 14 footprints were found in the mud

_____ No more than 7 keys can open the vault

2 **Write the letter of the matching item on each line.**

Match each detective clue phrase to its inequality symbol.

_____	at least	(A)	≥
_____	at most	(B)	≤
_____	more than	(C)	>
_____	fewer than	(D)	<
_____	no more than	(E)	≤

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A clue says the suspect carried at least 12 stolen gems. Which inequality shows the number g of gems?

(A) $g \geq 12$

(B) $g > 12$

(C) $g \leq 12$

(D) $g < 12$

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A witness reports that fewer than 5 people entered the building. Which inequality shows the number n of people?

(A) $n < 5$

(B) $n \leq 5$

(C) $n > 5$

(D) $n \geq 5$

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which inequality represents: 'A number n is at least 14'?

(A) $n \geq 14$

(B) $n > 14$

(C) $n \leq 14$

(D) $n < 14$

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Level 2 Extension — Catch the Symbol Mix-Up

1. **Clue** The suspect is at least 12 years on the force, where y is the years.
2. **Find the key phrase** 'At least 12' means 12 is allowed and anything larger.
3. **Write the inequality** y > 12

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do the words 'at least' and 'at most' change the inequality symbol compared to 'more than' and 'fewer than'?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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